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Chapter 1: INTRODUCTION

1.1 Reason for report

Bidding, or procurement, is a fundamental process in both public and private economies.This is a process where organizations-called inviting party, invite incumbent contractors-called bidding party, who submit their proposals and bids to supply goods, services or carry out the project. Bidding processes help to choose out the contractor meeting the criterias while ensuring an equally competitive environment in the industry and affect positively to the public economy. Especially in the USA, the procurement process is regulated in detail by the Federal Acquisition Regulation (FAR) and is overseen by government organizations such as the Department of Justice and the Federal Trade Commission.

However, the purpose of bidding is ideally to ensure equal competition between companies but it is very vulnerable to fraud known as Bid rigging. According to the Federal Trade Commision, bid rigging has many forms but the frequent form is the collusion between the opponents to choose the winner in advance. Bid rigging affects the competitiveness and the economy significantly so it should be identified as soon as possible. But this type of criminal is quite challenging to identify. For example, in the case of the Commonwealth of Kentucky in its bid to supply half-pin containers of fluid milk products for its school districts. The products provided include whole white milk, low-fat white milk, and low-fat chocolate milk. This case will be analyzed in the following part of the assignment as an example of bid fraud.

1.2 Goal of this report

After being inspected by the Commonwealth of Kentucky, Meyer and Trauth dairies were alleged to have criminal collusion. While Meyer admitted to their behavior, Trauth still maintains their innocence. Based on the provided background information, data in ASS02_MILK.xlsx, this case study report will give the writer's analysis and aim to demonstrate collusion evidence of this market. This case study mentions a school district milk supply bidding investigation. According to the previous findings, two dairies in the Tri-County market were suspected of "price-fixing" and "allocating the winner". Therefore, this report will focus on examining the market through 5 key indicators of the collusive bidding market including Market share, Incumbency rates, Bid dispersion, Average winning bid prices and Price versus cost.

Additionally, the market structure will be assessed against key characteristics that facilitate collusion, including high market concentration, homogeneous products, inelastic demand, and similar cost structures. While these conditions create an environment conducive to collusion, further analysis of bid prices is necessary to determine the extent of competition in the market.

Chapter 2: ANALYTICAL METHODS AND RESULTS

2.1 Market Share

Market share is a key metric used to detect signs of bid rigging, especially in the market with limited competition. In economics, market share is defined as the percentage of total product sales in a market that is held by a specific company. It reflects both market power and competitiveness. In the procurement market, persistently stable or rotating market shares over time can be a strong indicator of collusion (Kwoka, 1978).

Market share is calculated using the following formula and rounded to two decimal places:

$$Market\ Share\ (\%) = \frac{Number\ of\ milk\ half-pints\ supplied\ by\ the\ dairy\ over\ a\ given\ school\ year}{Total\ number\ of\ milk\ half-pints\ supplied\ \ to\ the\ entire\ market}$$

Year	Meyer	Trauth
1983	90,30%	9,70%
1984	45,22%	54,78%
1985	48,79%	51,21%
1986	54,25%	45,75%
1987	51,13%	48,87%
1988	53,60%	46,40%
1989	32,36%	67,64%
1990	29,03%	70,97%
1991	45,90%	54,10%

Table 1: Data of market share of each firm in Tri-County market from 1983 to 1991

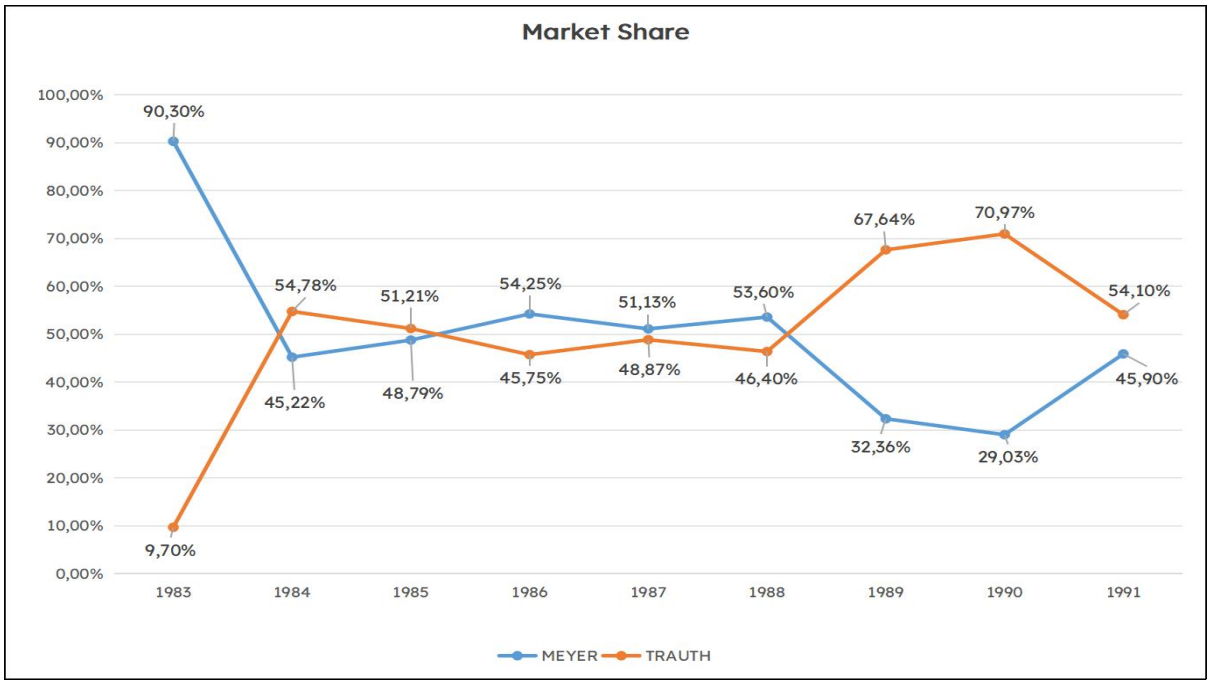

Figure 1: Market Share of each firm in Tri-County market from 1983 to 1991

Based on economic theories and previously identified collusive bidding patterns, market shares remain relatively stable or rotate predictably over many years is one sign of collusion. In a normally competitive market, changes in price, business strategies, and firm capabilities would significantly affect bidding outcomes. However, according to the chart, from 1984 to 1988, there were noticeable patterns between Meyer and Trauth, the market share was stable around 50%. When Meyer's market share decreased, Trauth's increased, and vice versa.

Moreover, in the healthy competitive market, any qualified company should have the opportunity to win contracts. Nevertheless, throughout the decade, contracts for milk distribution in the Tri-County market still belong to Meyer and Trauth. While other dairies were able to win contracts in surrounding markets but never succeeded in Tri-County, indicating potential market allocation. This occurred despite changes in production costs, consumer demand, and bidding policies so there is every reason to believe that something unusual is happening. This sign has been mentioned in "red flags of collusion" identified by the U.S. Department of Justice, which warns that:

"*When a small group of bidders controls most of the market and the product is homogeneous, the likelihood of collusion increases.*" (DOJ, 2024)"

2.2 Incumbency rate

Kwoka (1978) argues that "Market shares and persistence of winners over time are commonly used as indicators of potential collusive behavior in repeated bidding contexts."
In a bidding system, the incumbency rate is also a critical indicator used to detect signs of collusive behavior, especially in public tenders and large-scale procurement projects. This indicator exhibits the ability of suppliers in maintaining their market position over time and also signals the competitiveness and transparency in the bidding process.A high incumbency rate shows the incumbent's advantages in experience and service quality. On the other hand, a low rate demonstrates that the market has high competition or that the incumbents are not able to meet bidding criteria anymore. According to economic documents, the normal incumbency rate usually lay under 70%. If this rate consistently exceeds 70%, it raise red alert of bid rigging, such as firms rotating winning bids or allocating markets amongst themselves.

In this case study, based on the data from Milk.xlsx, the incumbency rate is calculated by using the following formula and rounded to two decimal places:

$$Incumbency\ rate = \frac{Number\ of\ districts\ award\ to\ the\ same\ supplier\ as\ previous\ years}{Total\ number\ of\ district\ in\ that\ year} x100$$

Year	Tri-County market			Surround market		
	Suspected win[1]	Total districts	Incumbency rates	Suspected win	Total districts	Incumbency rates
1983	0	11	0,00%	0	28	0,00%
1984	8	12	66,67%	16	28	57,14%
1985	12	13	92,31%	19	32	59,38%
1986	13	13	100,00%	19	37	51,35%
1987	12	13	92,31%	27	37	72,97%
1988	13	13	100,00%	25	37	67,57%
1989	9	13	69,23%	23	38	60,53%
1990	10	14	71,43%	24	35	68,57%
1991	12	13	92,31%	3	5	60,00%

Table 2: Incumbency rates data of both market from 1983 to 1991

[1] Number of districts award to same supplier as previous years

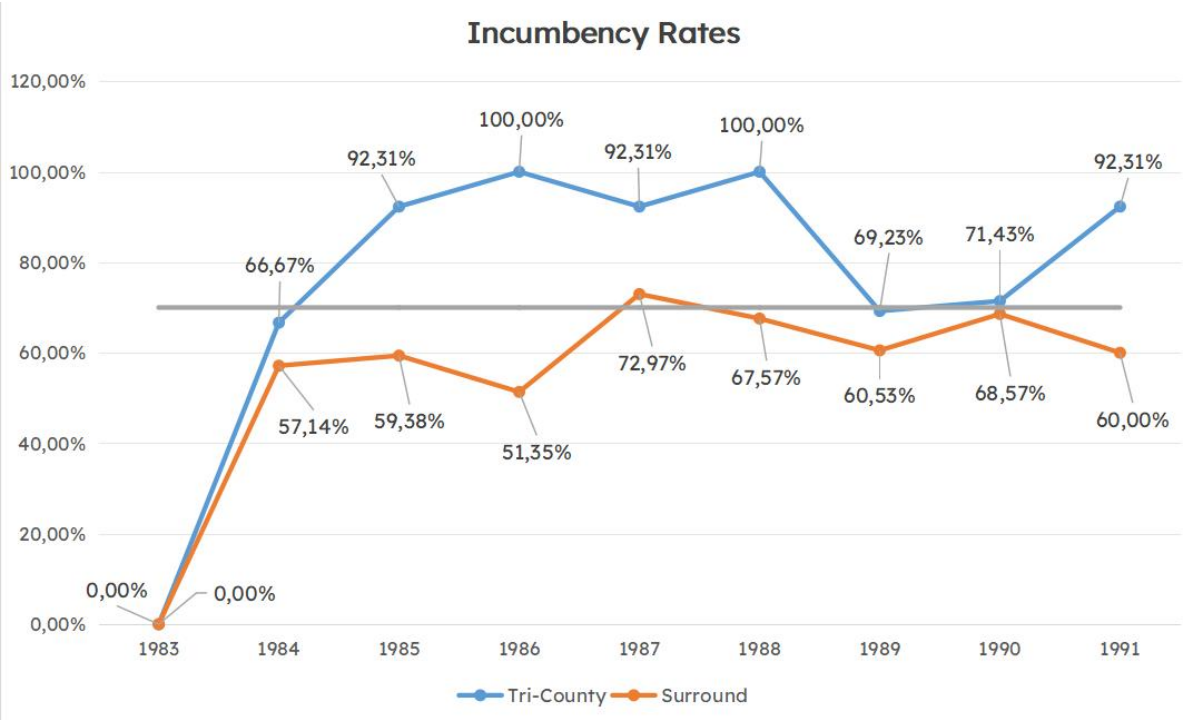

Figure 2: Incumbency rates of both market from 1983 to 1991

2.2.1 Tri-County Market Analysis

From the chart, Tri-County market displays a notably high and constant incumbency rate. Starting at 66.67% in 1984, the rate jumped to 92.31% in 1985 and then hit 100% in 1986 and 1988. Throughout the 1984–1991 period, the incumbency rate in this market remained above 90% in five out of seven years, rarely falling below 69%.

Stay consistency in rotating providers over a long period of time is unusual in a competitive public procurement market and is an indication of collusion among contractors. In a normal competitive market, it is natural to change contractors overtime. Therefore, in the Tri-County market, two suppliers, Meyer and Trauth, have maintained control of most of the district's contracts throughout that period.

This is consistent with economic warning signs of collusion, "A consistently high rate of repeat winners is one of the clearest signs of possible collusion."(Albano et al., 2006). As Kwoka (1978) noted, unusually high and stable incumbent rates, when combined with low turnover rates among winning firms, are consistent with market-sharing or bid-rotation schemes.

2.2.2 Surround Market Analysis

In contrast, Surround market shows more variation in incumbency rates, ranging between 51.35% (1986) and 72.97% (1987). Unlike Tri-County, this market shows regular turnover in contract owners, indicating high competition in this market.

Although the Surround market also had some peak points-such as nearly 73% in 1987-typically hovers close to the expected competitive benchmark of around 60–70%. This level of variability indicates a competitive and dynamic bidding environment.

Moreover, while Tri-County often had the same company winning contracts in a given year, this pattern was not observed in the Surround market, where contracts were more evenly rotated across multiple suppliers. This supports the notion that Surround experienced fewer fraudulence in the procurement process and was unlikely to cause collusion.

2.3 Bidding dispersion

In competitive auction markets, the standard deviation of winning bids serves as a measure of price dispersion and the level of competition. According to Harrington (2008), a low standard deviation over time can be a warning sign of price-fixing or collusion, as it suggests that prices are controlled by one or a few bidders and thus do not reflect true competition. On the other hand, a higher standard deviation indicates a healthy market where bidders offer more diverse prices due to independent cost structures and strategic behavior.
The formula used is:

$$Standard\ deviation = x = \sqrt{\frac{\Sigma(x_i - \bar{x})^2}{n-1}}$$

In that:
x_i: *winning bid prices*
$\bar{x}$: *mean winning bid prices*
n: *number of bids*

The analysis below will go deep into standard deviation of each product's categories: Whole white milk, Low-fat white milk and Low-fat chocolate milk to find collusive signals in Tri-County-suspected market and Surround market from 1883 to 1991.

2.3.1 Whole White Milk

Years	1983	1984	1985	1986	1987	1988	1989	1990	1991
Tri-County	0,0142	0,0047	0,0053	0,0044	0,0053	0,0050	0,0094	0,0041	0,0046
Surround	0,0143	0,0137	0,0132	0,0110	0,0103	0,0114	0,0075	0,0079	0,0065

Table 3: Standard deviation data of Whole White Milk in Tri-County and Surround

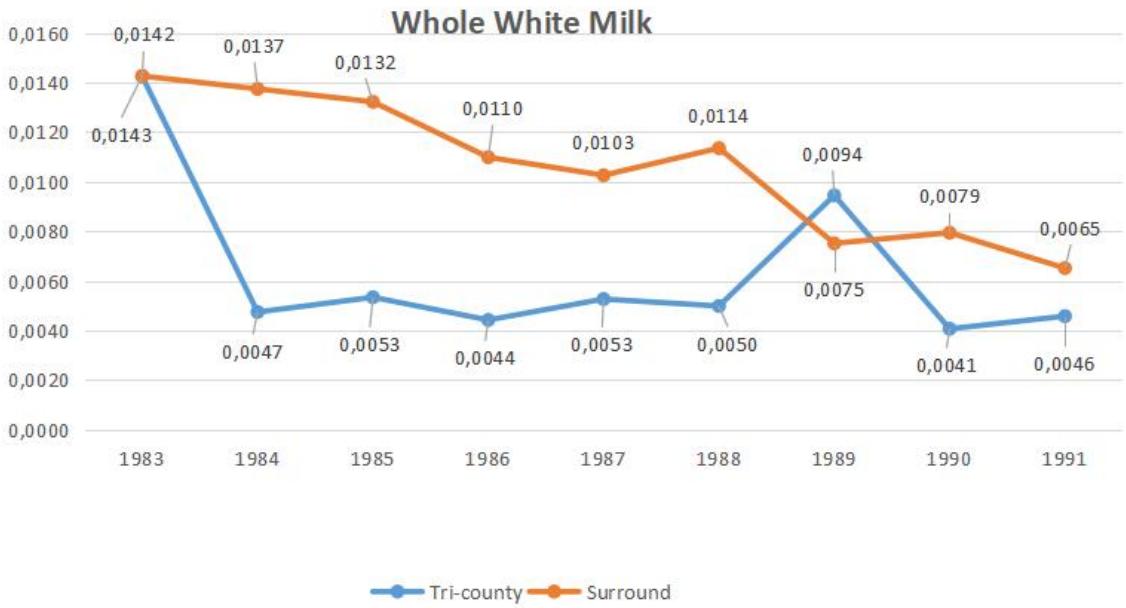

Figure 3: Standard deviation of Whole White Milk in Tri-County and Surround market

Based on the graph and table, Tri-County shows a significantly lower standard deviation value than Surround in most of the years. For example, in 1984, Tri-County had a standard deviation of 0,0047 while Surround had 0,0137-a relatively large gap and this continued until 1988. In 1989, Tri-County suddenly increased from 0,005 to 0,0094 but was still lower than 0,0075 of Surround. According to the U.S. Department of Justice (2013), “Be alert to situations in which all prices seem to be uniform

and all suppliers refuse to negotiate those prices." The continuous lower price change in Tri-County indicates the possibility of collusive bidding, as prices remain narrowly concentrated so let's continue to analyze the remaining two products

2.3.2 Low-fat White Milk

Years	1983	1984	1985	1986	1987	1988	1989	1990	1991
Tri-County	0,0122	0,0064	0,0054	0,0053	0,0070	0,0062	0,0083	0,0062	0,0059
Surround	0,0164	0,0147	0,0145	0,0118	0,0124	0,0129	0,0066	0,0095	0,0072

Table 4: Standard deviation data of Low-fat White Milk in Tri-County and Surround market

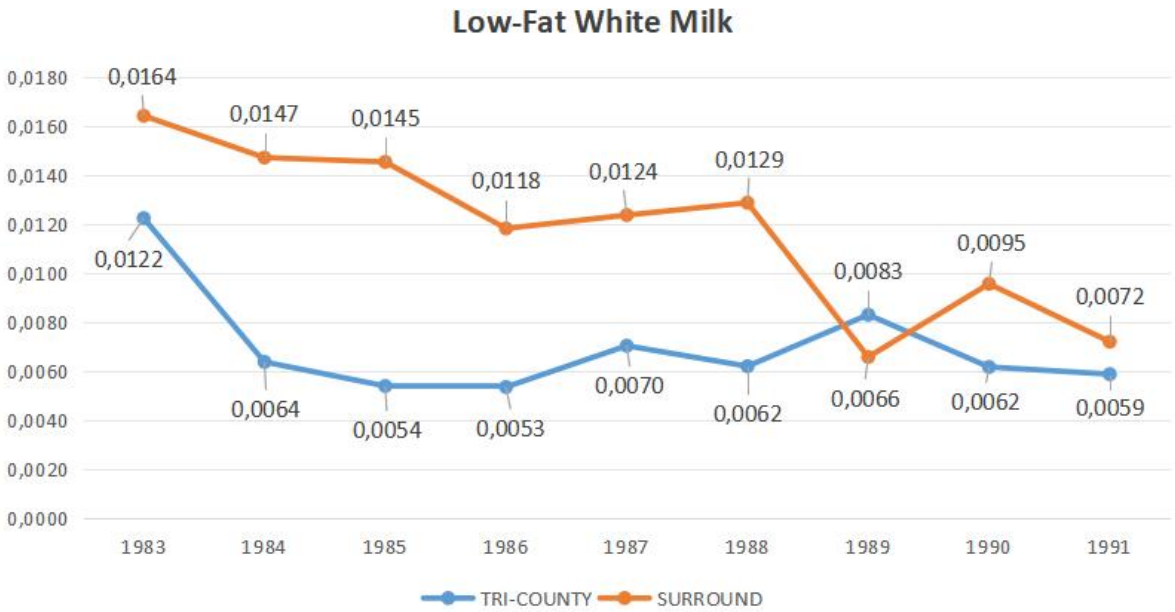

Figure 4: Standard deviation of Low-fat White Milk in Tri-County and Surround

As the chart shows, we can observe that for this product, the same phenomenon appears again. Tri-County market's standard deviation ranged from a low of 0,0053 to a high of 0,0070 during the period 1984-1988, while Surround market remained significantly higher. For example, 0,0145 in 1985 or 0,0129 in 1988 were far from Tri-County's 0,0054 and 0,0062, respectively. These differences again suggest that winning prices do not vary much, which is a sign of price fixing and collusion.

2.3.3 Low-fat Chocolate Milk

Years	1983	1984	1985	1986	1987	1988	1989	1990	1991
Tri-County	0,0039	0,0078	0,0062	0,0052	0,0063	0,0043	0,0093	0,0048	0,0065
Surround	0,0166	0,0153	0,0158	0,0128	0,0128	0,0136	0,0077	0,0099	0,0099

Table 5: Standard deviation data of Low-fat Chocolate Milk in Tri-County and Surround market

This product category shows the clearest signs of anomalies. In 1983, Surround had a standard deviation of 0,0166 while Tri-County had only 0,0039. Even as Tri-County increased slightly over the years (up to 0,0065 in 1991), Surround remained above 0,0099. The price variation between the two markets remains quite large.

2.4 Average winning prices

Average winning bids are often used as a key indicator of potential collusion if they exhibit unusual stability and consistency across markets. In a competitive market, winning bids are expected to fluctuate due to varying costs, supplier strategies. In the absence of such signs of competition, especially in tenders for the supply of goods, price consistency can serve as a warning sign of anti-competitive behavior. In theory:

"According to theoretical literature, "the mean winning price in the 'rigged' market will be significantly higher than the mean price in the competitive market for each year in which collusion occurs"

The mean winning bid prices in each product will be calculated using the following formula:

$$Average\ winning\ prices\ (by\ year\ and\ product\ type) = \frac{\Sigma\ \text{Winning price}}{\Sigma\ \text{Bids in that year}}$$

2.4.1 Whole White Milk

Years	1983	1984	1985	1986	1987	1988	1989	1990	1991
Tri-county	0,1186	0,1389	0,1415	0,1446	0,1495	0,1513	0,1314	0,1620	0,1412
Surround	0,1317	0,1309	0,1279	0,1253	0,1264	0,1277	0,1297	0,1609	0,1452

Table 6: Average winning prices of Whole White Milk in Tri-County and Surround market

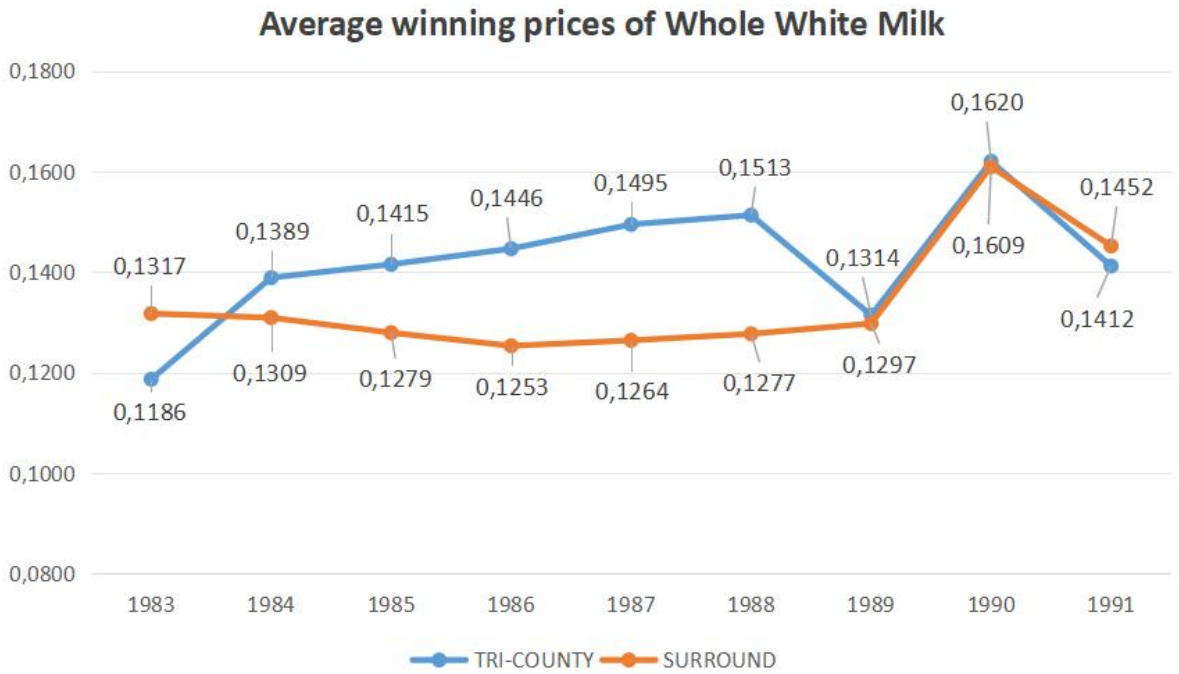

Figure 5: Average winning prices of Whole White Milk in Tri-County and Surround market

From 1983 to 1988, the Tri-County market consistently showed higher average winning bids than the Surrounding market. For example, 0,1513 compared to 0,1277 in 1988-quite a difference. However, during the 1989–1991 period, the numbers gradually closed in, with the Surrounding market sometimes even higher than the Tri-County market. For example, in 1991, the Surrounding market had a value of 0,1452 compared to the Tri-County market of 0,1412.

2.4.2 Low-fat White Milk

Years	1983	1984	1985	1986	1987	1988	1989	1990	1991
Tri-county	0,1121	0,1338	0,1366	0,1391	0,1424	0,1448	0,1240	0,1560	0,1354
Surround	0,1245	0,1236	0,1200	0,1178	0,1173	0,1182	0,1185	0,1519	0,1364

Table 7: Average winning prices of Low-fat White Milk in Tri-County and Surround market

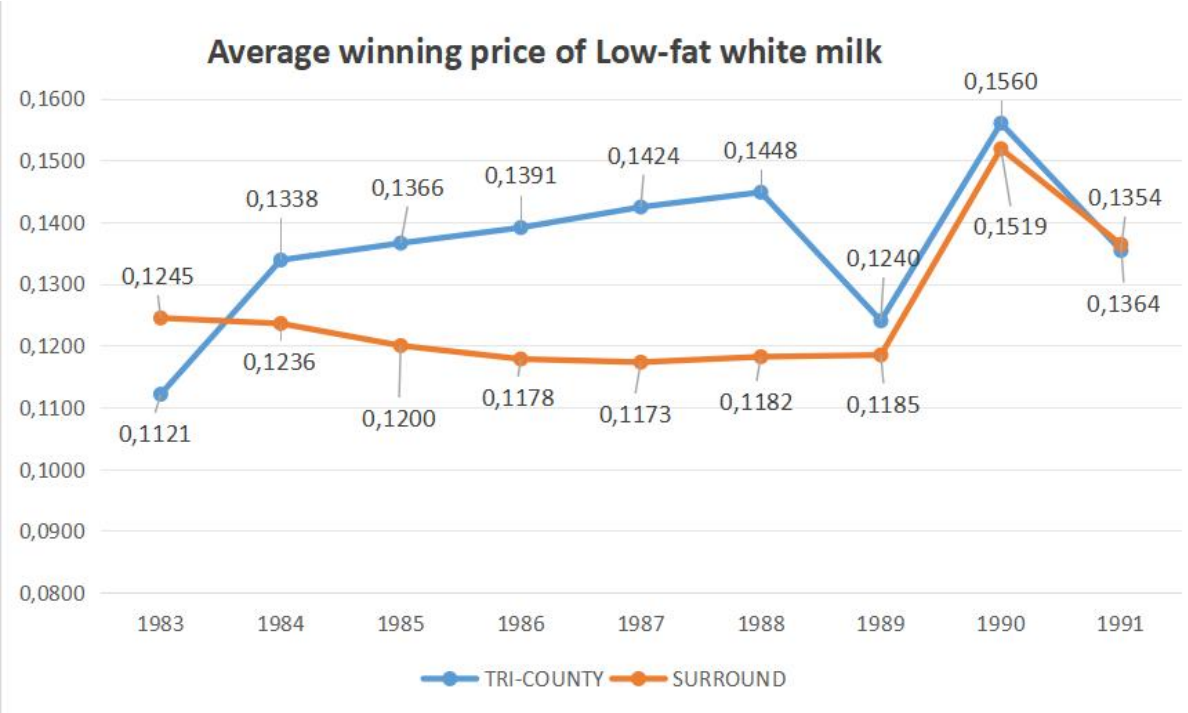

Figure 6: Average winning prices of Low-fat White Milk in Tri-County and Surround market

Similarly, the Tri-County market again showed significantly higher average prices between 1985 and 1988. In 1986, the Tri-County market was 0,1391 compared to the Surround market of 0,1178. The difference narrows in later years, but outliers remain.

2.4.3 Low-fat Chocolate Milk

Years	1983	1984	1985	1986	1987	1988	1989	1990	1991
Tri-county	0,1060	0,1347	0,1387	0,1414	0,1436	0,1470	0,1258	0,1605	0,1423
Surround	0,1266	0,1251	0,1206	0,1169	0,1184	0,1192	0,1199	0,1531	0,1402

Table 8: Average winning prices of Low-fat Chocolate Milk in Tri-County and Surround market

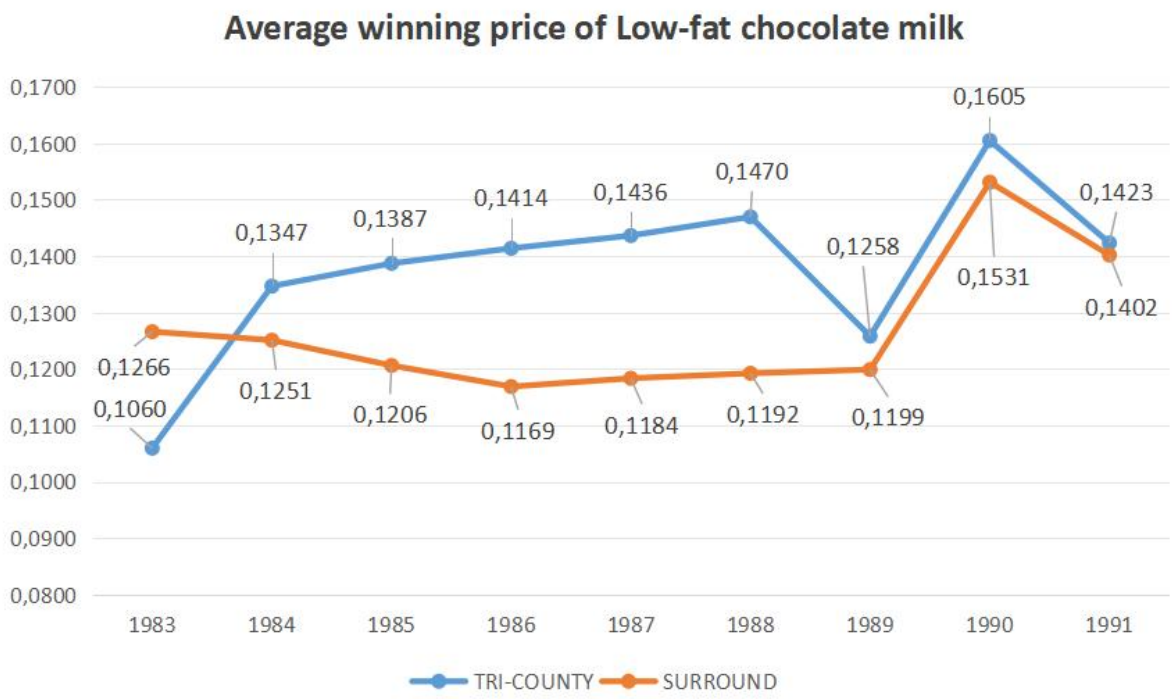

Figure 7: Average winning prices of Low-fat Chocolate Milk in Tri-County and Surround market

This product shows the most notable difference. During the 1985–1988 period, Tri-County's average price exceeded Surround's by a significant margin, in 1987, Tri-County was 0,1436 compared to Surround's 0,1184. From 1989, the average prices began to close in on the two products.

2.5 Prices versus costs

In a competitive bidding market, milk bid prices typically reflect production costs, especially raw milk costs, which account for about 60% of a dairy company's total costs. In this section, we will use the data to examine the correlation between average winning bid prices and raw milk costs (KYFMO) to assess whether prices in the Tri-County and Surround markets exhibit signs of competitive anomalies.

To investigate this relationship, we calculate the Pearson correlation coefficient between average winning bid prices and milk costs for each region. This coefficient indicates the strength and direction of the linear relationship between two continuous variables:

$$r = \frac{\Sigma(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\Sigma(x_i - \bar{x})^2(y_i - \bar{y})^2}}$$

For faster and easier calculation, data will be used in Excel scatter plot chart for faster calculation.

Market	Tri-County	Surround
Correlation coefficient (r)	0.049276984	0.513789975

Table 9: Correlation coefficient of Tri-County and Surround market

2.5.1 Surround Market

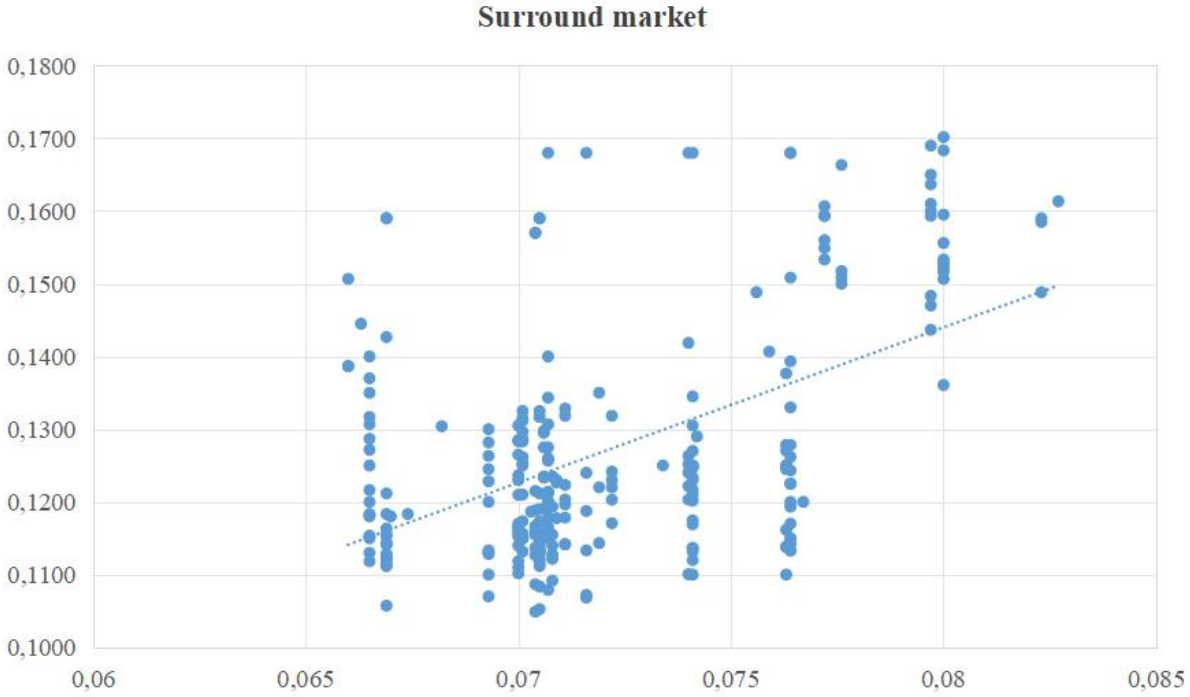

Figure 8: Correlation coeffiient of Surround market

This moderate positive correlation indicates that in the Surround market, as milk prices increase, winning bids tend to increase. This relationship is normal in a healthy competitive market where firms bid to match rising costs in order to maintain profitability. The scatter plot above shows this trend. Although the data points appear to be spread out, the upward trendline confirms the overall positive relationship between costs and prices.

2.5.2 Tri-County Market

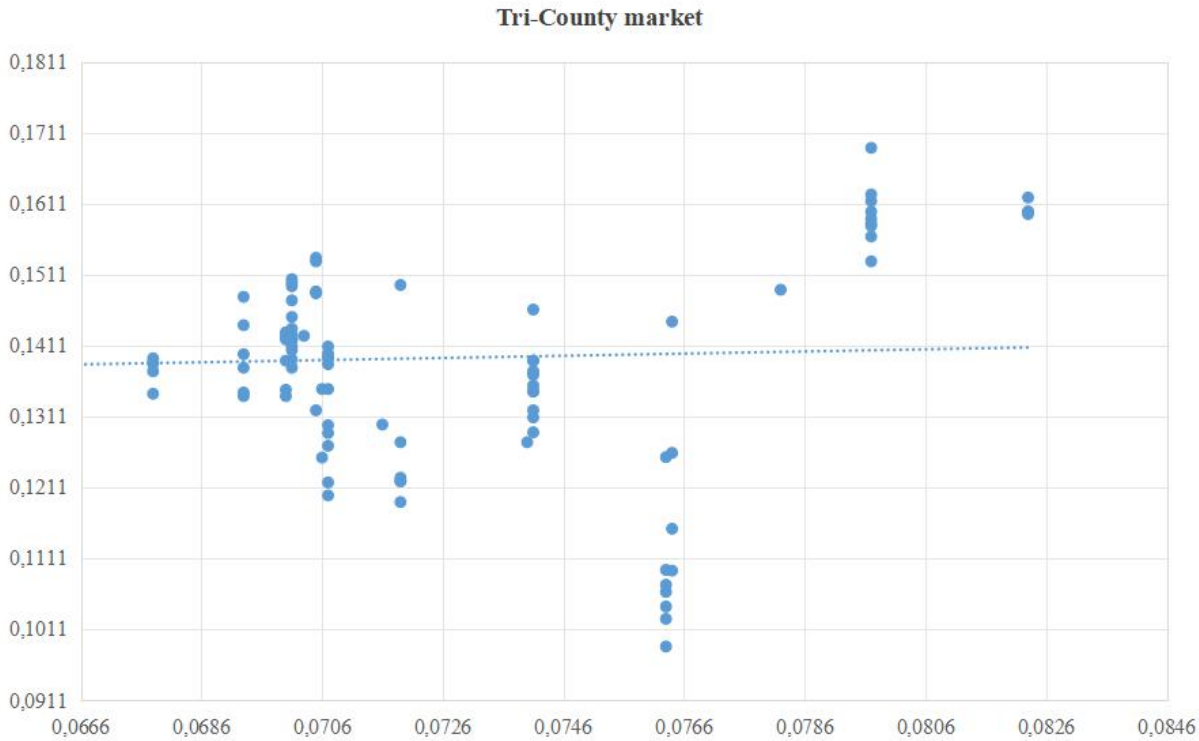

Figure 9: Correlation coeffiient of Tri-County market

On the other hand, the Tri-County market shows a very weak correlation between price and cost. Correlation coefficient (r) nearly equal zero demonstrate that price changes are not affected much by changes in milk costs in this market.

These findings are supported by the scatter plot chart, which shows a nearly flat trend line. Despite the variations in milk costs, the winning bids remain relatively around the same level. This indicates that firms seem not try to be competing to gain a cost advantage. This is a suspicious sign of unfair competitive behavior, such as price fixing or collusion.

Chapter 3 : CONCLUSION

Analysis of five key indicators from tasks 1 to 5, including Market share, Incumbency rate, Bid dispersion, Average winning bid price, and Correlation between price and cost in both Tri-County and Surround market. Compared to the surrounding market, Tri-County shown higher average prices, lower price volatility, higher incumbency rate, and significantly stable results, all consistent with established indicators of bid rigging. These findings strongly suggest that competition in Tri-County may have been controlled by a few individuals, undermining the integrity of the procurement process.

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